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Experiment No. 10

Aim: Evidence extraction from live system and Malware analysis

Title: To extract and analyse data collected from RAM

Part A:

Capturing and analysing volatile data is essential for discovering important evidence. Making a RAM dump should become a standard operating procedure when acquiring digital evidence before pulling the plug and taking the hard drive out. Live Acquisition involves the capture of data from a system that is running when you encounter it.

When to consider live capture:

- Loss of data during shutdown
- Pagefile set in registry to wipe at shutdown.
- “Evidence eliminator” apps that remove data at shutdown.
- Data not stored on disk (RAM contents, open ports, running processes, logged on users, etc.)
- Encryption
- Full Disk Encryption or open encrypted volumes
- Cached passwords/passphrases in RAM
- Volume of Data
- Too much to image everything?
- If you don’t need it all...
- Incident Response
- Volatile data lost if you turn off the computer.
- Suspect processes running only in RAM, not on disk.
- Court or client-imposed business interruption restrictions
- Kiosk/Internet Café
- Maybe no hard drive, booted by CD and everything is in RAM.

Types of Evidence Available in Volatile Memory

Many types of evidence are available in computer’s volatile memory that can be extracted by analysing memory dumps. Volatile and ephemeral evidence types include:

- Running processes and services.
- Unpacked/decrypted versions of protected programs.
- System information (e.g., time elapsed since last reboot)
- Information about logged in users.
- Registry information
- Open network connections and ARP cache
- Remnants of chats, communications in social networks and MMORPG games
- Recent Web browsing activities including IE InPrivate mode and similar privacy-oriented modes in other Web browsers.
- Recent communications via Webmail systems
- Information from cloud services
- Decryption keys for encrypted volumes mounted at the time of the capture.
- Recently viewed images.
- Running malware/Trojans.

Tools for Memory Forensics

Memory forensics is the analysis of volatile data in a computer's memory dump. It is conducted by many information security professionals to examine and identify attacks or malicious behaviours that do not leave readily detectable tracks on hard drive data. Moreover, it includes investigation of advanced computer attacks which are stealthy enough to avoid leaving data on the computer's hard drive.

Here are some favourite memory forensics tools and framework that can aid you in conducting effective memory analysis and forensics:

Volatility

Volatility is the memory forensics framework. It is used for incident response and malware analysis. With this tool, you can extract information from running processes, network sockets, network connection, DLLs and registry hives. It also has support for extracting information from Windows crash dump files and hibernation files.

WindowsSCOPE

WindowsSCOPE is another memory forensics and reverse engineering tool used for analysing volatile memory. It is primarily used for reverse engineering of malware. It provides the capability of analysing the Windows kernel, drivers, DLLs, virtual and physical memory.

Mandiant RedLine

Mandiant RedLine is a favourite tool for memory and file analysis. It collects information about running processes on a host, drivers from memory and gathers other data like metadata, registry data, tasks, services, network information and Internet history to build a proper report.

HELIX3

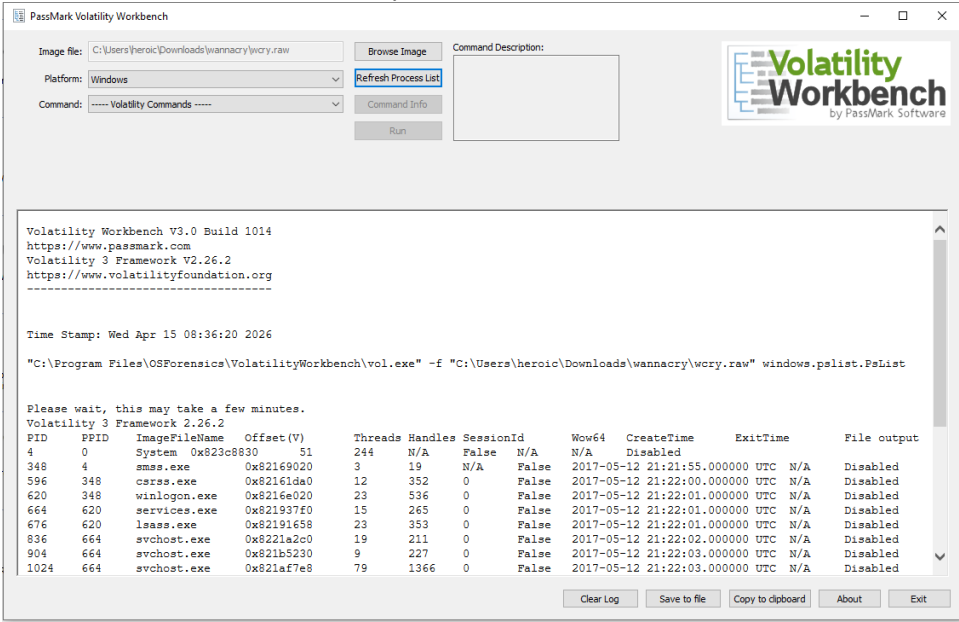
HELIX3 is a live CD-based digital forensic suite created to be used in incident response. This tool can collect data from physical memory, network connections, user accounts, executing processes and services, scheduled jobs, Windows Registry, chat logs, screen captures, SAM files, applications, drivers, environment variables and Internet history. Then it analyses and reviews the data to generate the compiled results based on reports.

Real Life Forensics

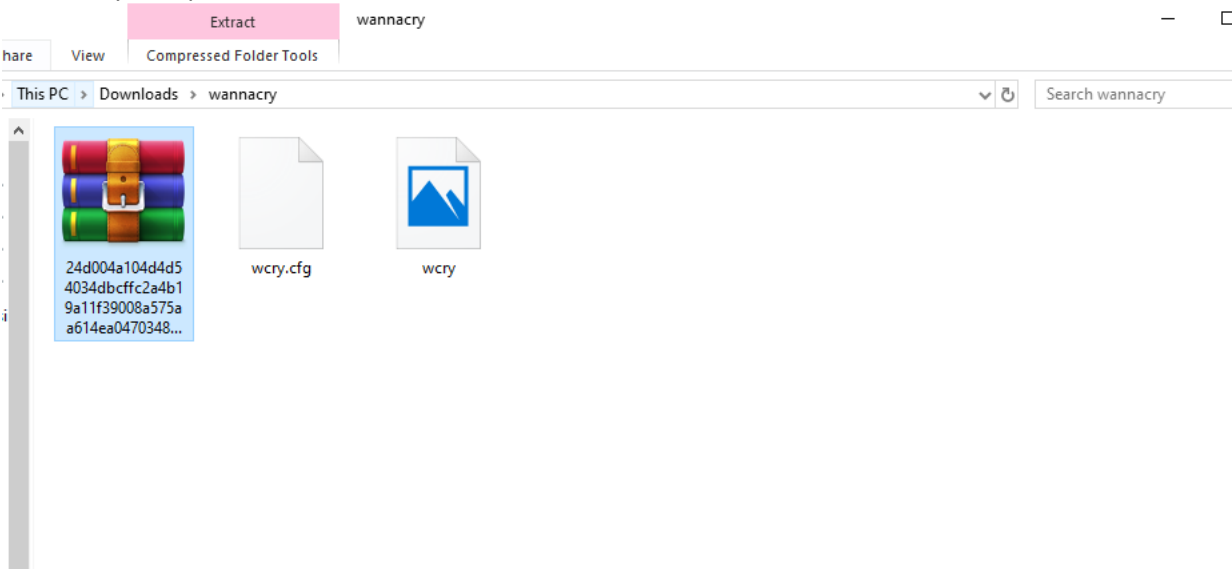
There are many reported cases of computer hacking that are prevented by applying computer forensics techniques and analysing the victim's machine. On one occasion Marquis-Boire and University of Toronto colleague Bill Marczak analysed e-mails received by Bahraini activists and found a piece of spyware designed to steal information from their computers. Further studies on the spyware showed similarities with the FinFisher surveillance software that Gamma International sells to law enforcement agencies.

This shows that how significantly the computer forensics is being used to prevent major incidents and analysing incidents to reveal its origin. It will help security teams to ban such traffic and contacts that are sending malicious and infected files.

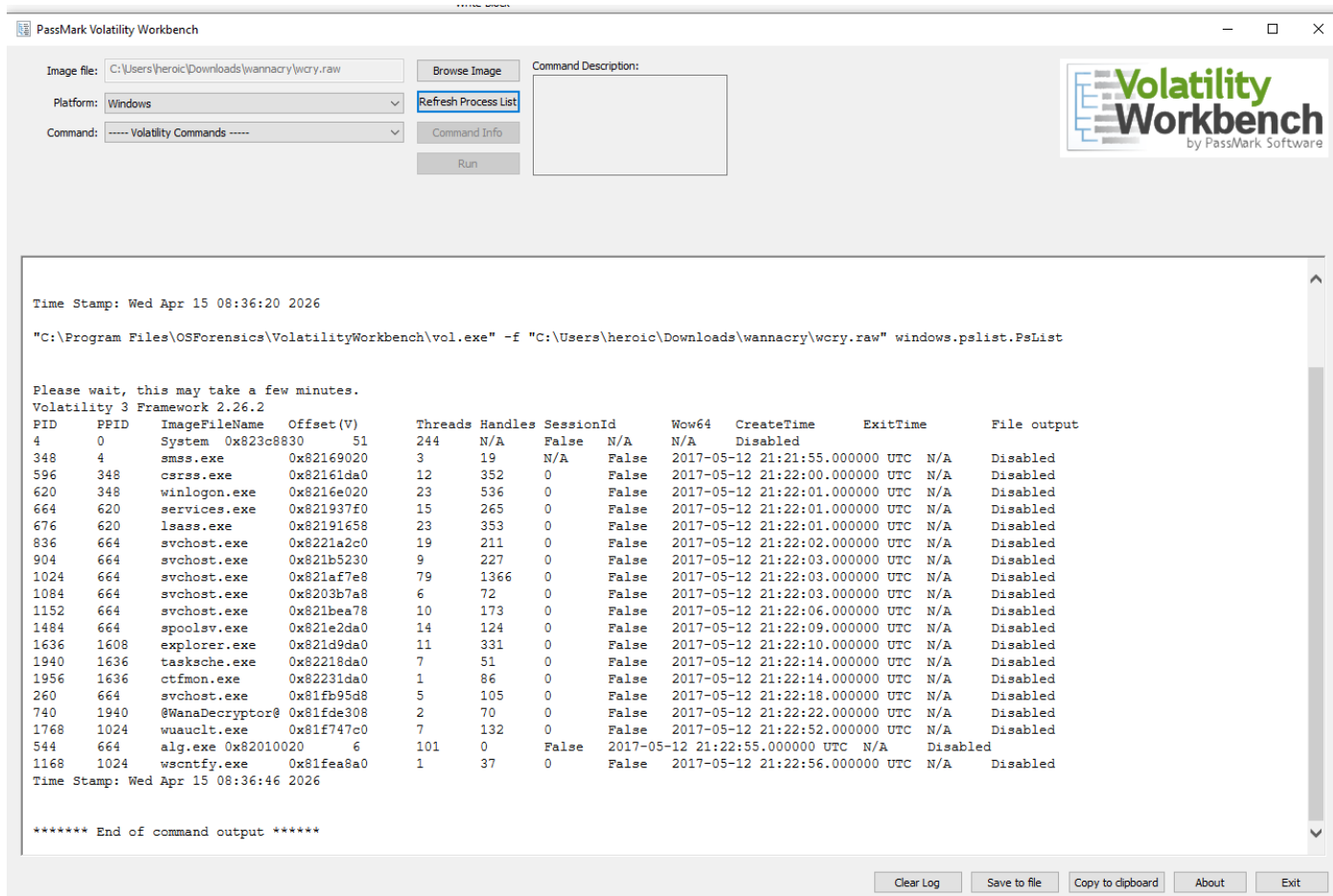
OSforensics also has the Volatility GUI.



WannaCry dump



wannacry ransomware attack



If we analyze the memory dump, we can identify **@WanaDecryptor@**, which is the WannaCry ransomware process. This executable is responsible for encrypting files, displaying the ransom note, and managing the overall infection.

Key observations include the following:

- It has its own Process ID (PID 740).

- It is running as a user-level process (Session 0 in this case).

- It appears alongside normal processes, indicating that it was actively running at the time the memory was captured.

We can trace how it started by examining the Parent Process ID (PPID). The process **explorer.exe** often launches user-triggered programs. If **@WanaDecryptor@** has a parent such as **explorer.exe**, it suggests that the infection may have originated from user execution, such as clicking a malicious file or opening a phishing attachment. This helps analysts determine the infection vector.

Conclusion: Thus, in this way, we can analyze memory dumps from primary memory. Tools like Volatility provide the capability to understand the properties and behavior of running processes.